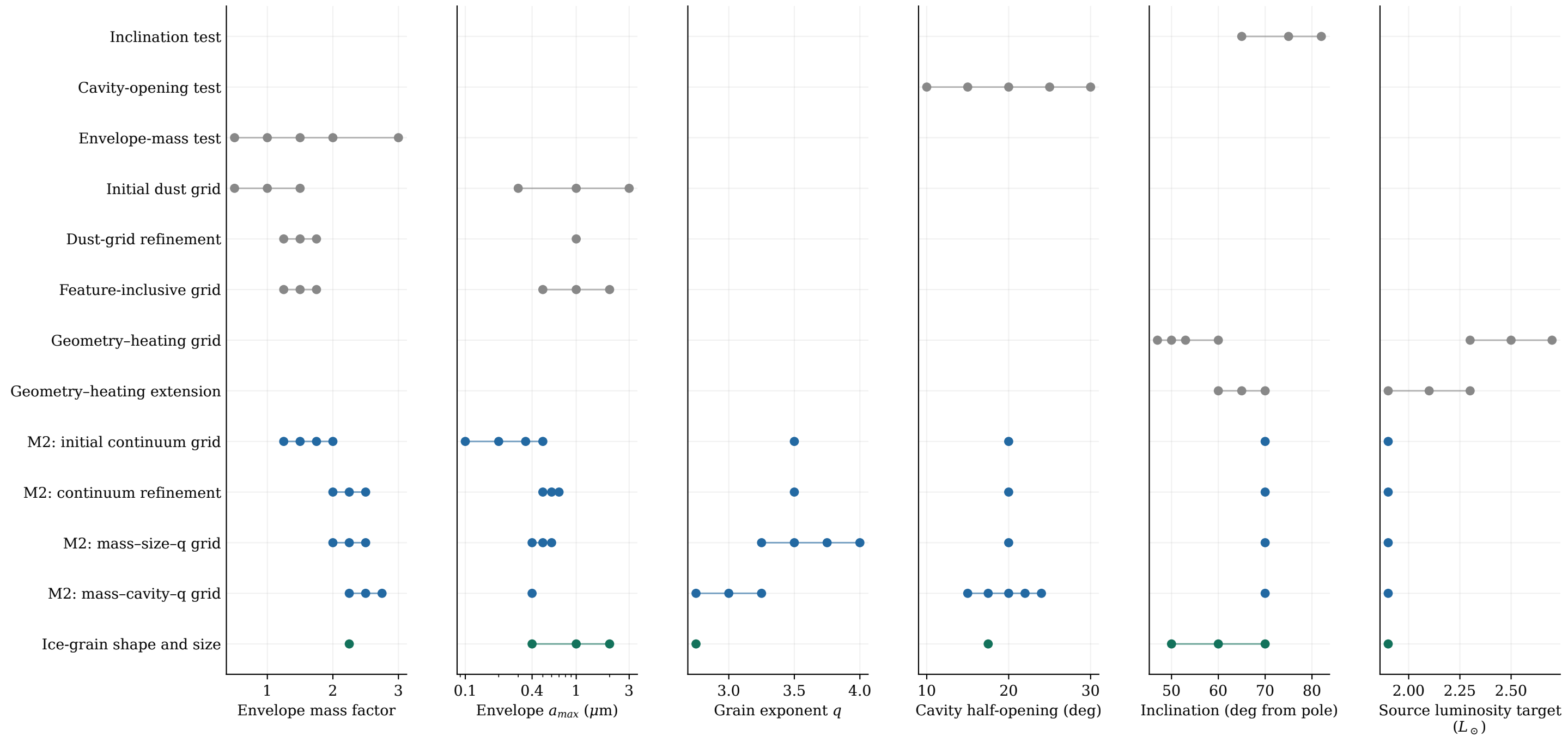
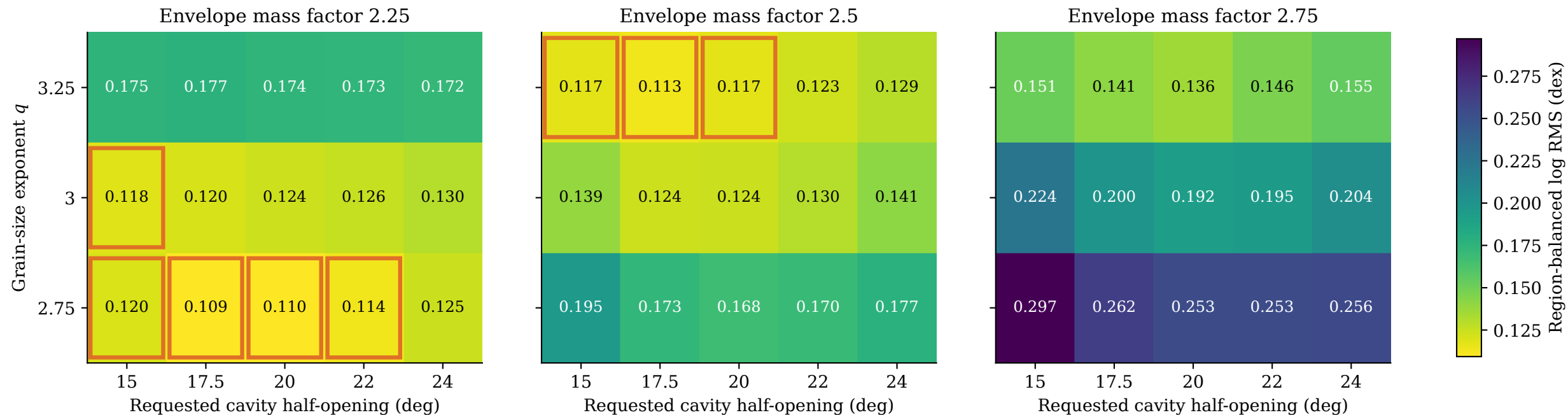


Sampled parameter space



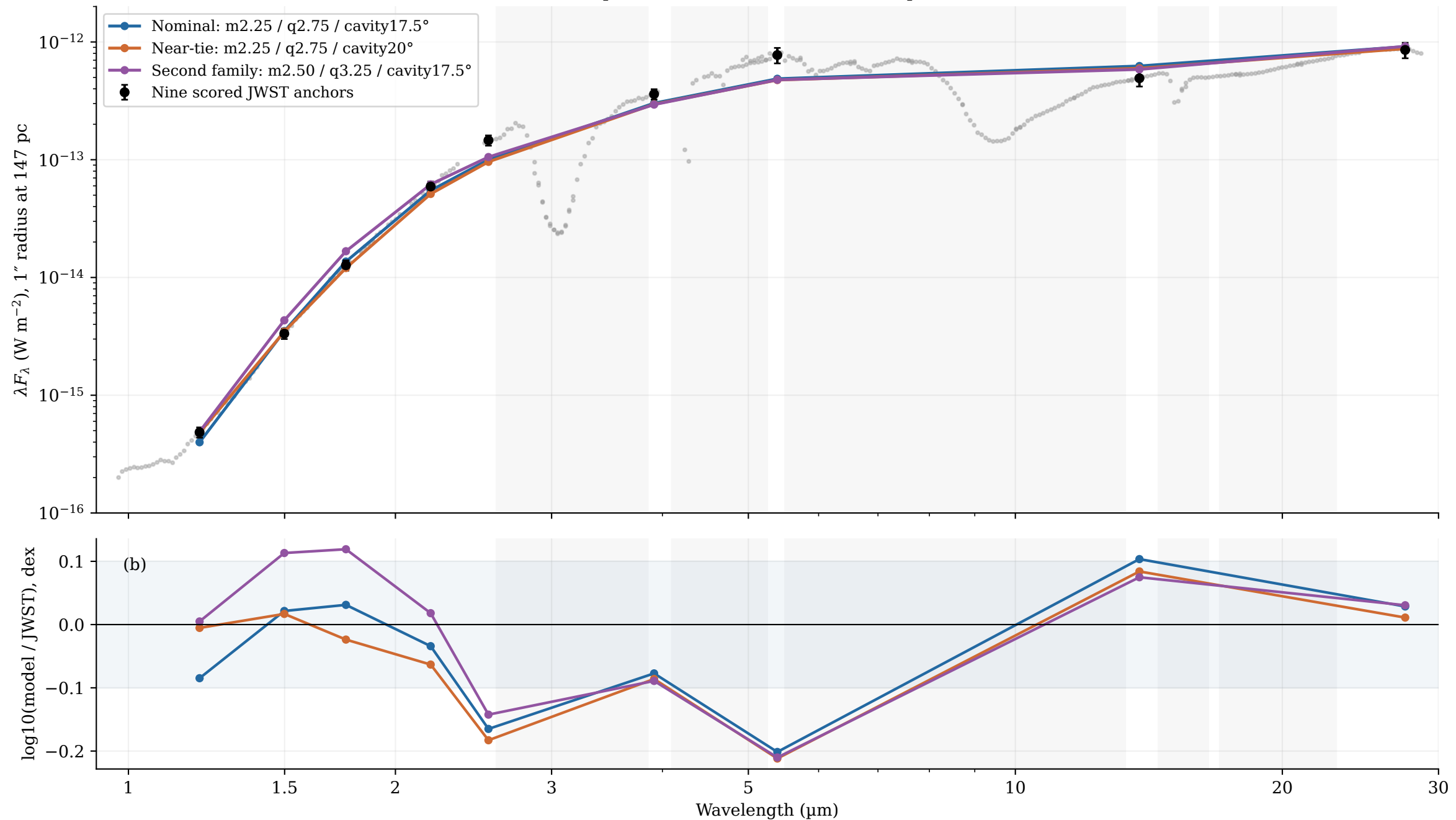
Grey: earlier estimators Blue: Method-2 (M2) continuum grids Green: ice-grain calculations
 Symbols denote sampled values. Lines indicate the sampled range, not continuous coverage. Fixed values are included where documented.

Continuum score distribution: mass-cavity-grain-exponent grid



Orange boxes: 8 models within the operational 0.01086-dex tolerance (not confidence intervals).
 Nominal best: 2.25 / 17.5° / $q=2.75$, score 0.10931 dex. Cavity 20° differs by only 0.00045 dex.
 Fixed: $i=70^\circ$, $a_{\text{max}}=0.4 \mu\text{m}$, source target 1.9 L_{sun} , original Mie grains with 5% ice mantle by volume.

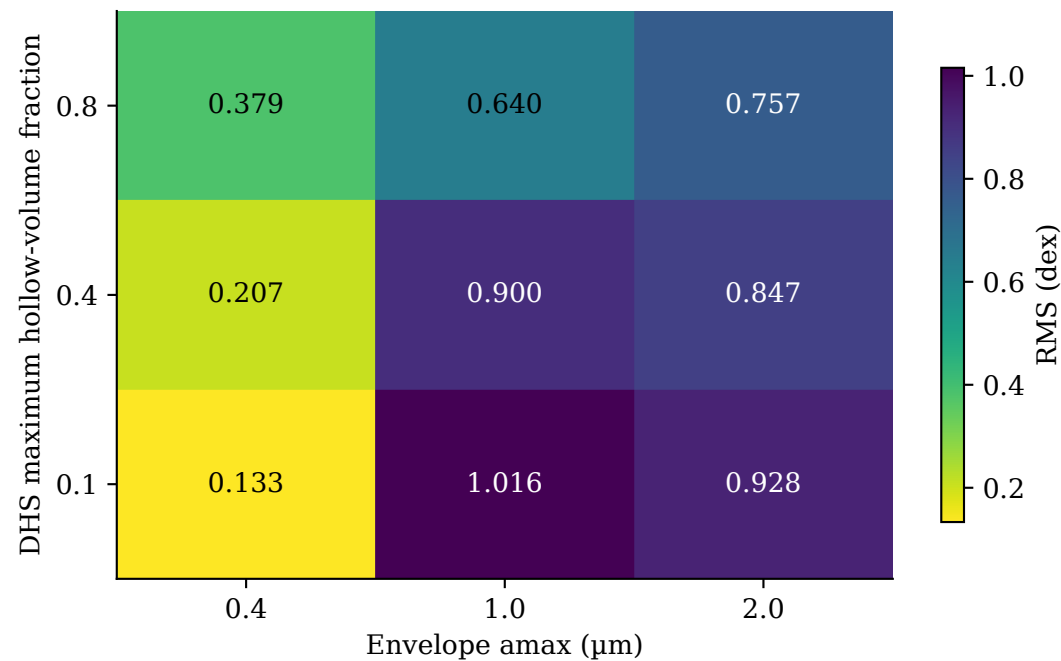
(a) Aperture-matched continuum comparison



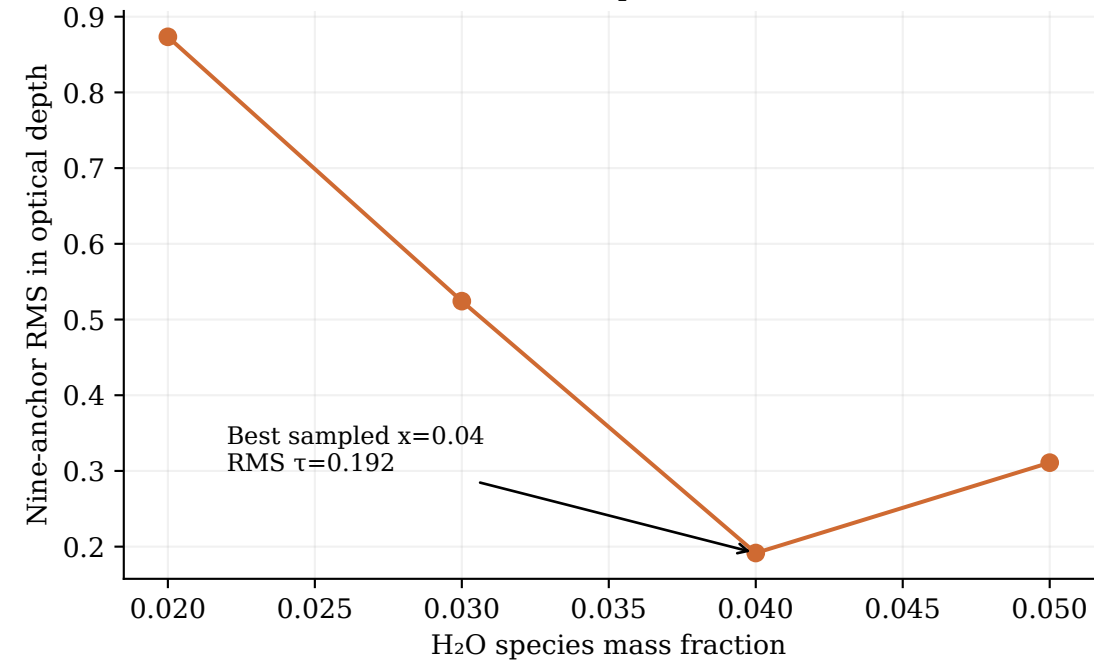
Grey points are observed $R \approx 100$ bins, including the excluded features. Connecting model segments are guides, not predictions inside those features. The ≈ 0.995 - μ m anchor is excluded. The score is a regional log-RMS screening statistic, not reduced chi-square or a posterior.

Continuum and absorption-band response to dust properties

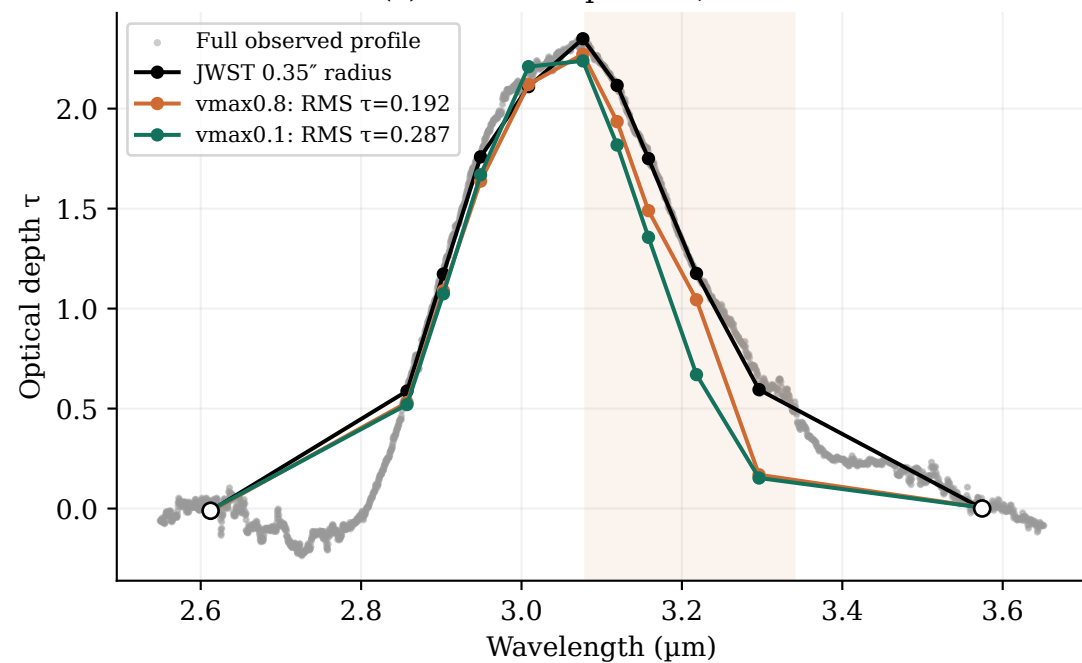
(a) Near-infrared continuum residuals, $i=70^\circ$



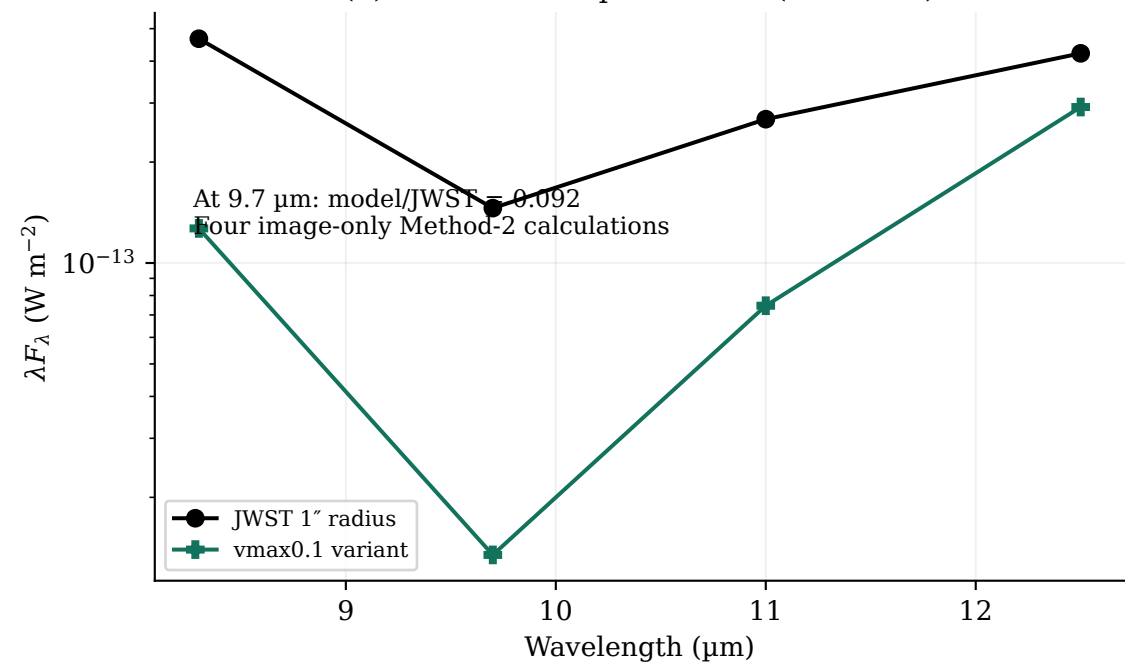
(b) H_2O abundance dependence, $v_{\text{max}}=0.8$



(c) H_2O -band profiles, $x=0.04$

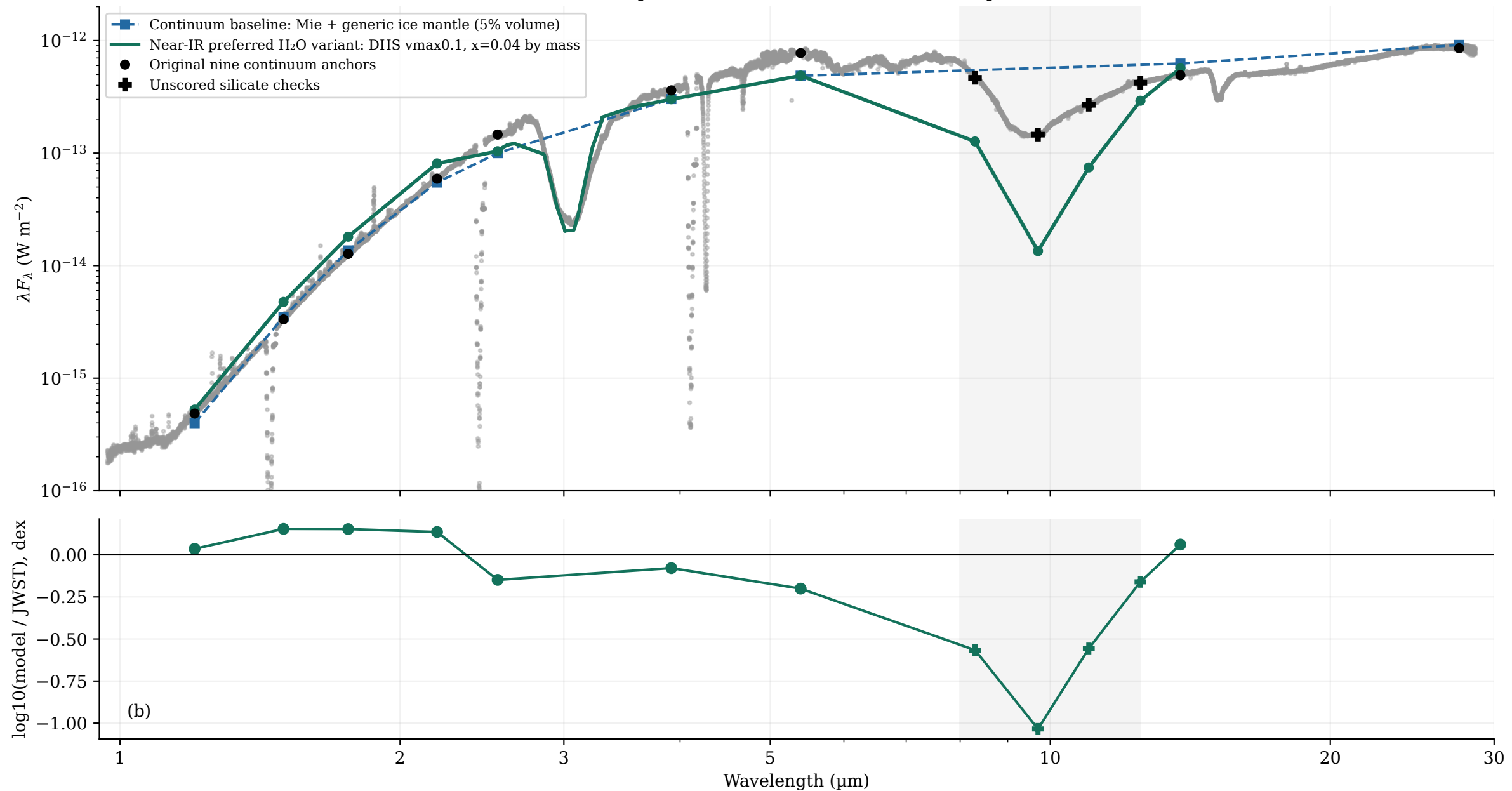


(d) Silicate-band predictions (not fitted)



Changed-dust results reuse the old continuum temperature: these are diagnostic screens, not a self-consistent global fit. Original continuum dust already contained 5% ice by mantle volume; the later H_2O fraction is a species mass fraction.

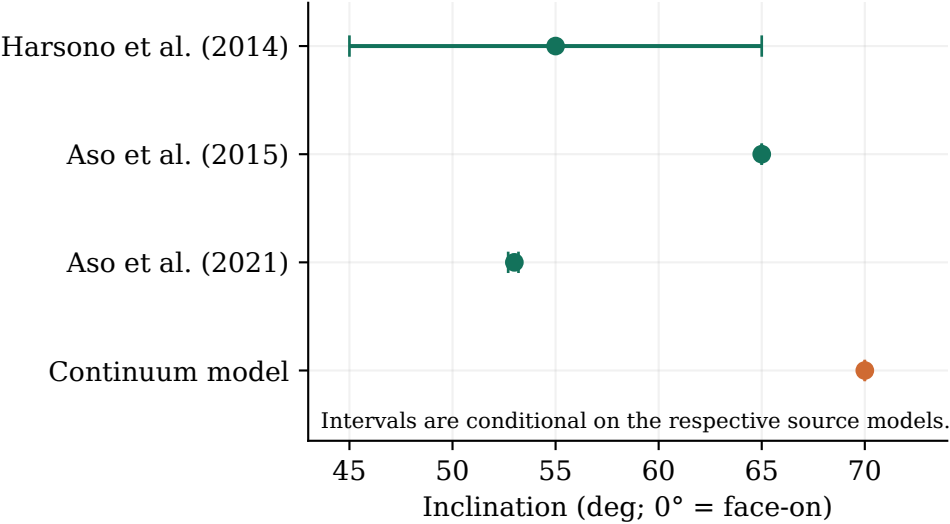
(a) Observed spectrum and conditional dust-model predictions



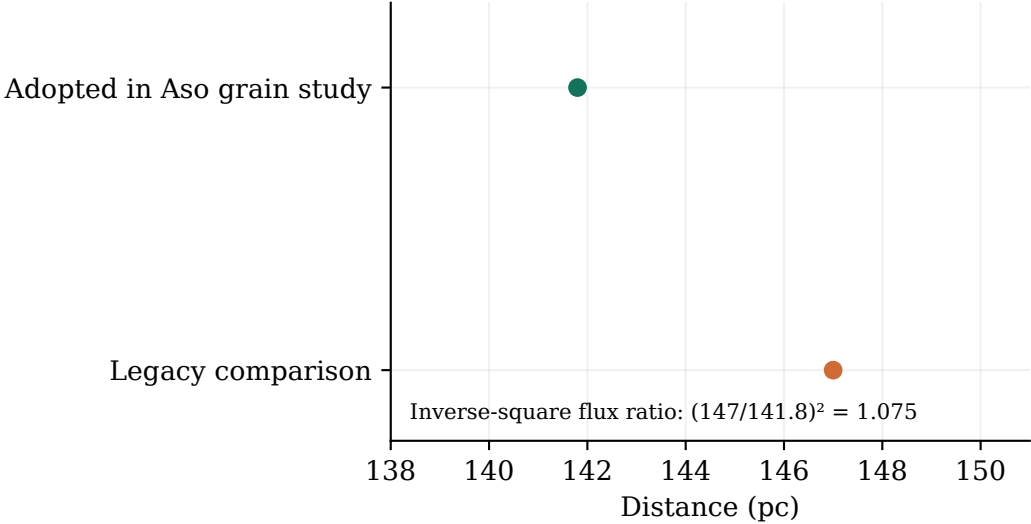
Aperture radius: 1". Grey points: native JWST spectrum. Connecting lines do not predict unsampled wavelengths.
H₂O optical-depth scores use a separate 0.35" aperture. No updated 27.5- μm prediction exists for the DHS variant.

Comparison with published source constraints

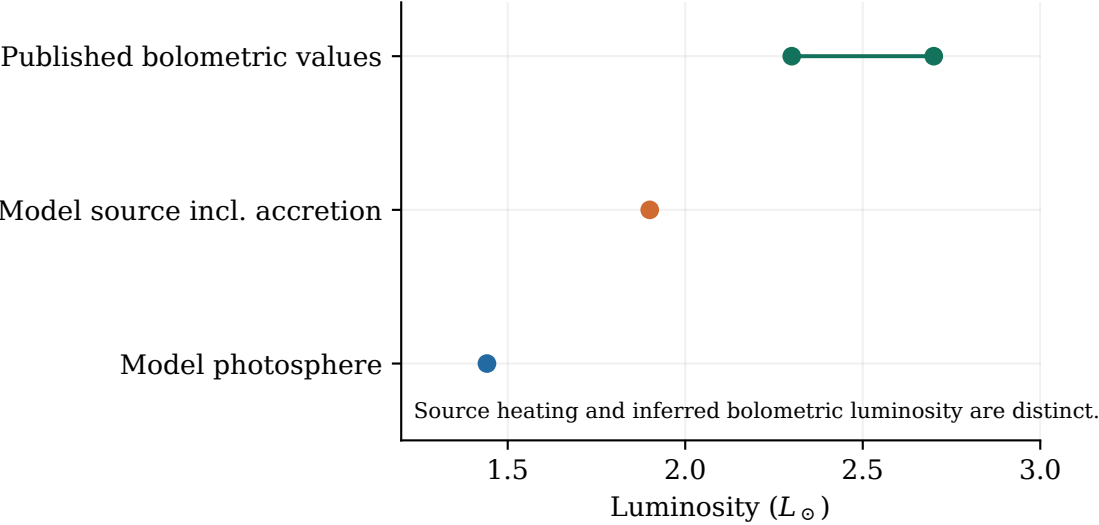
(a) Published inclination estimates



(b) Adopted source distances



(c) Luminosity definitions

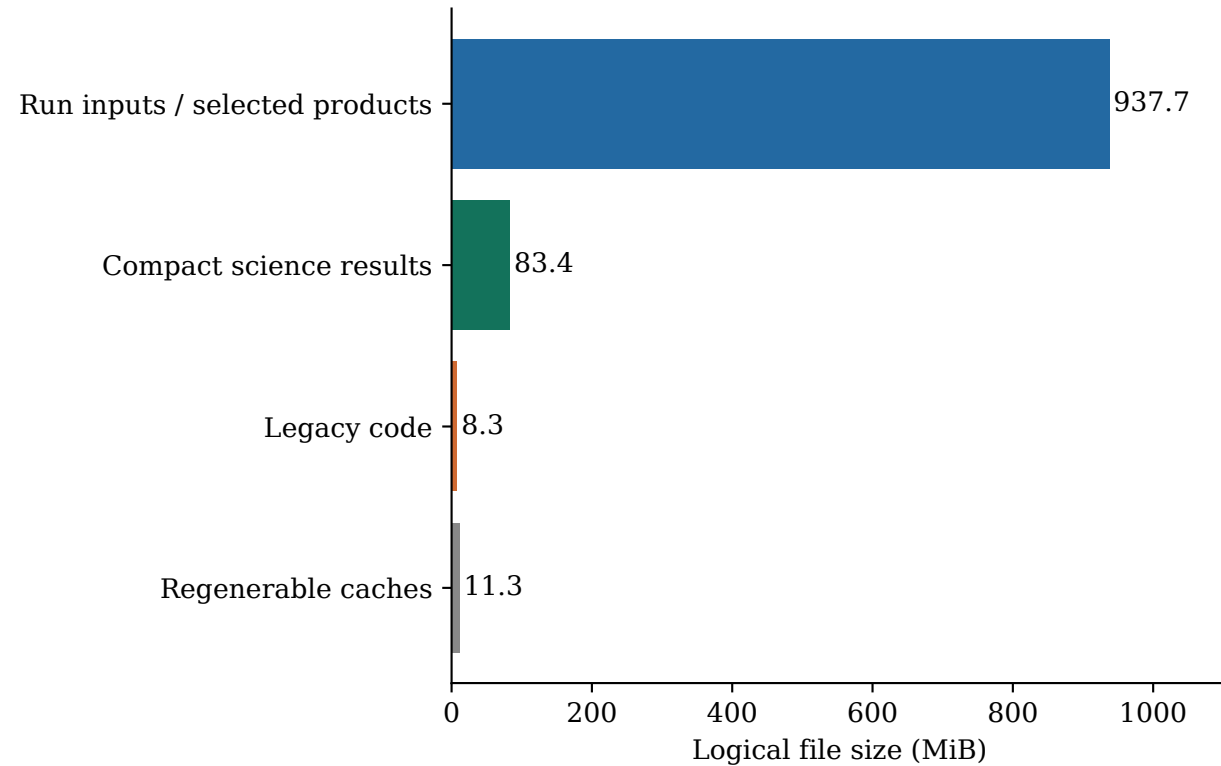


(d) Scope and applicability of published constraints

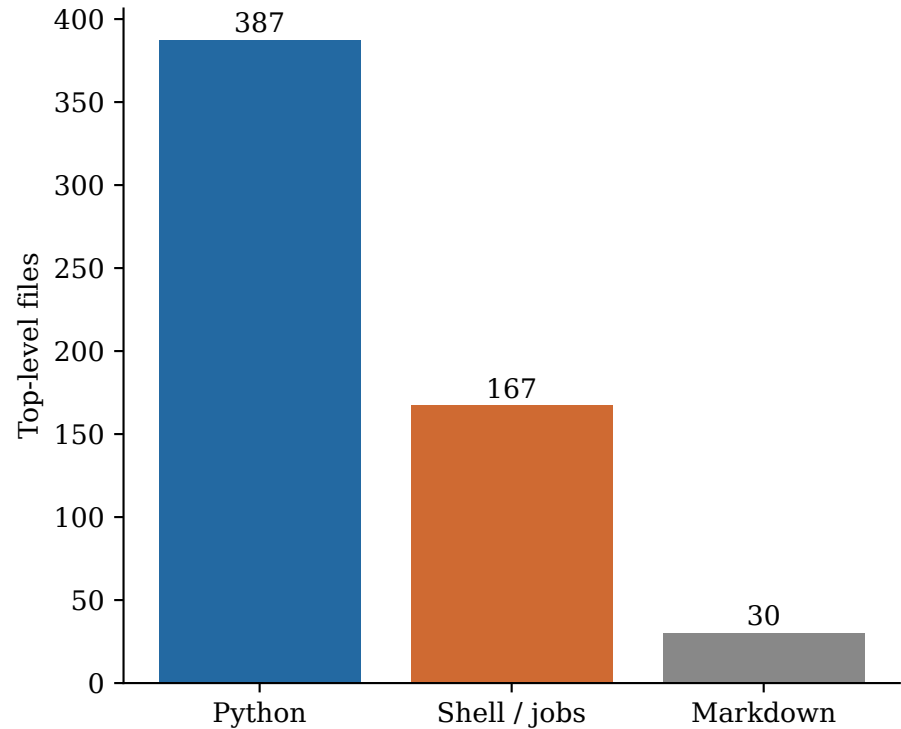
- Envelope grain size:**
The $\geq 10\text{-}\mu\text{m}$ radiative-alignment argument is not a source-specific lower limit.
- Extinction:**
Jet $AV \approx 17\text{--}20$ and apparent silicate $AV \approx 33$ sample different sightlines and assumptions.
- Grain populations:**
Disk and scattering-envelope constraints probe distinct grain populations.
- Mass and cavity:**
Comparisons depend on opacity, gas/dust ratio, spatial extent and full/half-angle convention.

References and measurement definitions are provided in the text. The displayed estimates do not constitute a combined posterior.

(a) Archived data volume by category



(b) Historical source-file inventory



Before housekeeping: 3,930 files, 1.02 GiB logical size. 267 source files resolve resources relative to their own location. Historical code and scientific products are retained in a verified archive with their original relative paths.